

Call for Expressions of Interest

<i>Date of Issue:</i>	September 2026
<i>Expected Completion Date:</i>	May 2027
<i>Project Title:</i>	Post-Quantum Cryptography for CriticalInfrastructure network security
<i>Company:</i>	Technigel Consulting
Contact Details	
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Additional Info	
<i>Students must sign NDA:</i>	☐
<i>Project uses personal data:</i>	☐
<i>Custom hardware required:</i>	☒ If Applicable: Raspberry Pi x 2.
<i>Relevant research areas (if applicable)¹:</i>	Secure networks, post quantumcryptography, cybersecurity (Cycos)
Project Background Information	
Critical-infrastructure organisations (energy, water, transport, utilities) depend on long-lifecycle operational technology (OT) devices that remain deployed for 10–25 years. These devices typically connect to HQ systems through VPN or SD-WAN overlays for telemetry, maintenance, and remote management. Today these tunnels rely on classical cryptography (RSA, DH, ECC), all of which are vulnerable to quantum attacks such as Shor’s algorithm. This creates a “harvest-now, decrypt-later” risk where rogue actors can capture encrypted traffic today and decrypt it once quantum capability becomes available. Industry bodies and regulators (including the UK NIS Regulations) now expect operators of essential services to plan for post-quantum cryptography (PQC) adoption. NIST has standardised PQC algorithms such as CRYSTALS-Kyber (key establishment) and CRYSTALS-Dilithium (signatures), and network equipment developers have started to explore PQ-enabled VPNs, PQ-TLS, and PQ-SDWAN prototypes. However, the practical impact of PQC on long-lifecycle, low-cost field devices, tunnel performance, and resilience against mid-line	

¹ An overview of different research areas in the school of computer science can be found [here](#).

interception is not yet well understood and is of concern to Chief Information and Security Officers across the industry.

This project addresses that gap by building a small-scale but realistic model of a critical-industry communication path using Raspberry Pi devices to represent long-lifecycle field equipment and a rogue actor with mid-line access.

Requirements

*The group must provide: a sandbox **three-node PQC communication model**:*

- 1. Headquarters System** A PQC-enabled VPN/SD-WAN initiator using Kyber/Dilithium to establish a quantum-safe secure tunnel.
 - 2. Field Device (Raspberry Pi 1)** A constrained endpoint terminating the PQC tunnel, representing a remote OT device, at a water treatment site
 - 3. Rogue Actor "Tap" (Raspberry Pi 2)** An inline interception node simulating an adversary with physical access to a network segment.
- The project will deliver three core work packages:*

(a) PQC-Enabled VPN/SD-WAN Testbed

Build and configure a working PQC-secured tunnel between HQ and the field device using PQC libraries (e.g., liboqs). Demonstrate existing and new secure communication aligned with NIS expectations for cryptographic robustness and secure remote access. Create a DNP3 (Distributed Network Protocol v3) over TCP payload traffic simulation appropriate to telemetry protocols currently in use.

(b) PQC Performance Benchmarking

Measure handshake latency, CPU load, memory usage, and throughput on the Raspberry Pi endpoint. Assess whether PQC adoption remains "appropriate and proportionate" for long-lifecycle critical-industry devices.

(c) Safe, Isolated PQC Incursion Simulation

Use the rogue actor Pi to perform controlled mid-line interception attempts (packet capture, replay, downgrade interference). Demonstrate how PQC mitigates harvest-now/decrypt-later threats and supports NIS incident-response readiness. All testing must be performed on isolated lab systems (sandbox) only. The final report and presentation must include a section that would advise the Chief Information and Security officer of a critical infrastructure organisation on a strategic approach.

Skills

Highly desirable:

- Basic Linux administration (Raspberry Pi OS, networking tools)
- Understanding of VPN or SD-WAN concepts
- Ability to configure and troubleshoot

Desirable:

- Experience with Raspberry Pi hardware
- Interest in post-quantum cryptography (Kyber, Dilithium)
- Familiarity with liboqs, OpenVPN,

<i>network tunnels</i> - Programming/scripting for automation and testing (Python, Bash) - Familiarity with cryptographic principles (keys, certificates, handshakes and provisioning) - Ability to perform benchmarking and interpret performance data	<i>WireGuard or TLS libraries</i> - Understanding of critical-infrastructure security and UK NIS Regulations - Experience with packet capture and analysis (tcpdump, Wireshark) - Ability to model threat scenarios and MITM techniques - Report writing and presentation skills
Submission / Award information	
<i>Date of Submission:</i>	September 2026
<i>Date of Pitch:</i>	October 2026
<i>Notification of award:</i>	October 2026

For further information on this proposal, the team representative should in the first instance e-mail the lead contact stated above at the e-mail address provided.